THE DETERMINANT OF A MATRIX

LINEAR ALGEBRA

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

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ENDTERM

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LONG QUIZ

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SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

The Determinant of a Matrix

- 1x1 Matrix
- 2 x 2 Matrix
- 3 x 3 Matrix

Laplace Expansion

- 4 x 4 Matrix

THE DETERMINANT OF A MATRIX

$$\boldsymbol{A} = [\boldsymbol{a}]$$

The determinant of a 1×1 matrix

$$\boldsymbol{det}(\boldsymbol{A}) = \boldsymbol{a}$$

Find the **determinant** of the matrix

$$A = [5]$$

$$\boxed{|A| = 5}$$

ans

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The determinant of a 2×2 matrix

$$det(A) = ad - bc$$

Find the **determinant** of the matrix

$$A = \begin{bmatrix} 3 & -2 \\ 4 & 5 \end{bmatrix}$$

Find the **determinant** of the matrix

$$A = \begin{bmatrix} 3 & -2 \\ 4 & 5 \end{bmatrix}$$

$$|A| = (3 \cdot 5) - (4 \cdot -2)$$

$$|A| = 15 + 8$$

$$\boxed{|A| = 23}$$

ans

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

The determinant of a 3×3 matrix

$$det(A) = aei + bfg + cdh$$
$$-gec - hfa - idb$$

Find the **determinant** of the matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 5 & 3 \\ 1 & -4 & 2 \\ 2 & 1 & -1 \end{bmatrix}$$

Find the **determinant** of the matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 5 & 3 \\ 1 & -4 & 2 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 5 & 3 \\ 1 & -4 & 2 \\ 2 & 1 & -1 \end{bmatrix}$$

$$|A| = (2 \cdot -4 \cdot -1) + (5 \cdot 2 \cdot 2) + (3 \cdot 1 \cdot 1)$$

$$-(2 \cdot -4 \cdot 3) - (1 \cdot 2 \cdot 2) - (-1 \cdot 1 \cdot 5)$$

$$|A| = (8) + (20) + (3)$$

$$-(-24) - (4) - (-5)$$

$$|A| = 31 + 25$$

$$\boxed{|A| = 56}$$

ans

LAPLACE EXPANSION

then the $\boldsymbol{ij}$ **minors** denoted by $\boldsymbol{M_{ij}}$ are

$$\boldsymbol{M_{11}} = \begin{vmatrix} \boldsymbol{a_{22}} & \boldsymbol{a_{23}} \\ \boldsymbol{a_{32}} & \boldsymbol{a_{33}} \end{vmatrix}$$

If A is a 3×3 matrix

$$\mathbf{A} = \begin{bmatrix} \boldsymbol{a_{11}} & \boldsymbol{a_{12}} & \boldsymbol{a_{13}} \\ \boldsymbol{a_{21}} & \boldsymbol{a_{22}} & \boldsymbol{a_{23}} \\ \boldsymbol{a_{31}} & \boldsymbol{a_{32}} & \boldsymbol{a_{33}} \end{bmatrix}$$

then the $\boldsymbol{ij}$ **minors** denoted by $\boldsymbol{M_{ij}}$ are

If A is a 3×3 matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

$$M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

then the $\boldsymbol{ij}$ **minors** denoted by $\boldsymbol{M_{ij}}$ are

If A is a 3×3 matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

$$M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

$$M_{13} = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

then the $\boldsymbol{ij}$ **minors** denoted by $\boldsymbol{M_{ij}}$ are

If A is a 3×3 matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

$$M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

$$M_{13} = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Determine the ij minors M_{11}, M_{12}, M_{13} of the matrix

$$A = \begin{bmatrix} 2 & 5 & 3 \\ 1 & -4 & 2 \\ 2 & 1 & -1 \end{bmatrix}$$

$$M_{11} = \begin{vmatrix} -4 & 2 \\ 1 & -1 \end{vmatrix}$$

$$M_{11} = (-4 \cdot -1) - (1 \cdot 2)$$

$$M_{11} = 4 - 2$$

$$\boxed{M_{11} = 2}$$

ans

$$M_{12} = \begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix}$$

$$M_{12} = (1 \cdot -1) - (2 \cdot 2)$$

$$M_{12} = -1 - 4$$

$$\boxed{M_{12} = -5}$$

ans

$$M_{13} = \begin{vmatrix} 1 & -4 \\ 2 & 1 \end{vmatrix}$$

$$M_{13} = (1 \cdot 1) - (2 \cdot -4)$$

$$M_{13} = 1 - (-8)$$

$$\boxed{M_{13} = 9}$$

ans

then the **cofactor** of element a_{ij} denoted by $\boldsymbol{C_{ij}}$ is

$$\boldsymbol{C_{ij} = (-1)^{i+j} M_{ij}}$$

If A is a 3×3 matrix

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Cofactor of A,

$$\begin{bmatrix} +M_{11} & -M_{12} & +M_{13} \\ -M_{21} & +M_{22} & -M_{23} \\ +M_{31} & -M_{32} & +M_{33} \end{bmatrix}$$

$$det(A) = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$$

The determinant of a 3×3 matrix

$$\mathbf{A} = \begin{array}{c} \begin{array}{ccc} + & - & + \end{array} \\ \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \end{array}$$

where,

$$a_{ij} = entry$$

$$M_{ij} = minor$$

Find the **determinant** of the matrix

$$\begin{array}{ccc} + & - & + \end{array}$$

$$A = \begin{bmatrix} 2 & 5 & 3 \\ 1 & -4 & 2 \\ 2 & 1 & -1 \end{bmatrix}$$

$$M_{11} = \begin{vmatrix} -4 & 2 \\ 1 & -1 \end{vmatrix} = 2$$

$$M_{12} = \begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix} = -5$$

$$M_{13} = \begin{vmatrix} 1 & -4 \\ 2 & 1 \end{vmatrix} = 9$$

$$|A| = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$$

$$|A| = 2(2) - 5(-5) + 3(9)$$

$$|A| = 4 + 25 + 27$$

$$\boxed{|A| = 56}$$

ans

Find the **determinant** of the matrix

$$|A| = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13} - a_{14}M_{14}$$

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 3 & 1 \\ 4 & 2 & 0 & 1 \\ 2 & 1 & 1 & 2 \end{bmatrix}$$

Cofactor of A =

$$\begin{bmatrix} +M_{11} & -M_{12} & +M_{13} & -M_{14} \\ -M_{21} & +M_{22} & -M_{23} & +M_{24} \\ +M_{31} & -M_{32} & +M_{33} & -M_{34} \\ -M_{41} & +M_{42} & -M_{43} & +M_{44} \end{bmatrix}$$

Find the **determinant** of the matrix

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 3 & 1 \\ 4 & 2 & 0 & 1 \\ 2 & 1 & 1 & 2 \end{bmatrix}$$

Cofactor of A =

$$\begin{bmatrix} +M_{11} & -M_{12} & +M_{13} & -M_{14} \\ -M_{21} & +M_{22} & -M_{23} & +M_{24} \\ +M_{31} & -M_{32} & +M_{33} & -M_{34} \\ -M_{41} & +M_{42} & -M_{43} & +M_{44} \end{bmatrix}$$

$$|A| = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13} - a_{14}M_{14}$$

$$M_{11} = \begin{vmatrix} 1 & 3 & 1 \\ 2 & 0 & 1 \\ 1 & 1 & 2 \end{vmatrix} \begin{vmatrix} 1 & 3 & 1 \\ 2 & 0 & 1 \\ 1 & 1 & 2 \end{vmatrix}$$

$$M_{11} = (1 \cdot 0 \cdot 2) + (3 \cdot 1 \cdot 1) + (1 \cdot 2 \cdot 1)$$
$$-(1 \cdot 0 \cdot 1) - (1 \cdot 1 \cdot 1) - (2 \cdot 2 \cdot 3)$$

$$M_{11} = 0 + 3 + 2$$
$$-0 - 1 - 12$$

$$M_{11} = -8$$

Find the **determinant** of the matrix

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 3 & 1 \\ 4 & 2 & 0 & 1 \\ 2 & 1 & 1 & 2 \end{bmatrix}$$

$$|A| = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13} - a_{14}M_{14}$$

$$M_{12} = \begin{vmatrix} 0 & 3 & 1 \\ 4 & 0 & 1 \\ 2 & 1 & 2 \end{vmatrix} \begin{vmatrix} 0 & 3 & 1 \\ 4 & 0 & 1 \\ 2 & 1 & 2 \end{vmatrix}$$

$$M_{12} = (0 \cdot 0 \cdot 2) + (3 \cdot 1 \cdot 2) + (1 \cdot 4 \cdot 1)$$
$$-(2 \cdot 0 \cdot 1) - (1 \cdot 1 \cdot 0) - (2 \cdot 4 \cdot 3)$$

$$M_{12} = 0 + 6 + 4$$
$$-0 - 0 - 24$$

$$M_{12} = -14$$

Cofactor of A =

$$\begin{bmatrix} -8 & -M_{12} & +M_{13} & -M_{14} \\ -M_{21} & +M_{22} & -M_{23} & +M_{24} \\ +M_{31} & -M_{32} & +M_{33} & -M_{34} \\ -M_{41} & +M_{42} & -M_{43} & +M_{44} \end{bmatrix}$$

Find the **determinant** of the matrix

$$|A| = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13} - a_{14}M_{14}$$

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 3 & 1 \\ 4 & 2 & 0 & 1 \\ 2 & 1 & 1 & 2 \end{bmatrix}$$

$$M_{13} = \begin{vmatrix} 0 & 1 & 1 \\ 4 & 2 & 1 \\ 2 & 1 & 2 \end{vmatrix}$$

Cofactor of A =

$$\begin{bmatrix} -8 & -(-14) & +M_{13} & -M_{14} \\ -M_{21} & +M_{22} & -M_{23} & +M_{24} \\ +M_{31} & -M_{32} & +M_{33} & -M_{34} \\ -M_{41} & +M_{42} & -M_{43} & +M_{44} \end{bmatrix}$$

Find the **determinant** of the matrix

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 3 & 1 \\ 4 & 2 & 0 & 1 \\ 2 & 1 & 1 & 2 \end{bmatrix}$$

Cofactor of A =

$$\begin{bmatrix} -8 & -(-14) & +M_{13} & -M_{14} \\ -M_{21} & +M_{22} & -M_{23} & +M_{24} \\ +M_{31} & -M_{32} & +M_{33} & -M_{34} \\ -M_{41} & +M_{42} & -M_{43} & +M_{44} \end{bmatrix}$$

$$|A| = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13} - a_{14}M_{14}$$

$$M_{14} = \begin{vmatrix} 0 & 1 & 3 \\ 4 & 2 & 0 \\ 2 & 1 & 1 \end{vmatrix} \begin{vmatrix} 0 & 1 & 3 \\ 4 & 2 & 0 \\ 2 & 1 & 1 \end{vmatrix}$$

$$M_{14} = (0 \cdot 2 \cdot 1) + (1 \cdot 0 \cdot 2) + (3 \cdot 4 \cdot 1)$$
$$-(2 \cdot 2 \cdot 3) - (1 \cdot 0 \cdot 0) - (1 \cdot 4 \cdot 1)$$

$$M_{14} = 0 + 0 + 12$$
$$-12 - 0 - 4$$

$$M_{14} = -4$$

Find the **determinant** of the matrix

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 3 & 1 \\ 4 & 2 & 0 & 1 \\ 2 & 1 & 1 & 2 \end{bmatrix}$$

Cofactor of A =

$$\begin{bmatrix} -8 & -(-14) & +M_{13} & -(-4) \\ -M_{21} & +M_{22} & -M_{23} & +M_{24} \\ +M_{31} & -M_{32} & +M_{33} & -M_{34} \\ -M_{41} & +M_{42} & -M_{43} & +M_{44} \end{bmatrix}$$

$$|A| = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13} - a_{14}M_{14}$$

$$|A| = 1(-8) - 2(-14) + 0(M_{13}) - 3(-4)$$

$$|A| = -8 + 28 + 0 + 12$$

$$\boxed{|A| = 32}$$

ans

SEATWORK